Linux Commands

◆ BASIC COMMANDS

whoami

Syntax: `whoami`

Purpose: Displays the username of the current logged-in user.

ls

Syntax: `ls`

Purpose: Lists files and directories in current folder.

cd

Syntax: `cd [directory]`

Example: `cd /etc`

Purpose: Changes directory.

pwd

Syntax: `pwd`

Purpose: Shows full path of current directory.

clear

Syntax: `clear`

Purpose: Clears terminal screen.

cat

Syntax: `cat [file]`

Example: `cat interfaces`

Purpose: Displays file content in terminal.

cp

Syntax: `cp source destination`

Example: `sudo cp ls backup_ls`

Purpose: Copies files.

sudo

Syntax: `sudo command`

Purpose: Runs command as root (admin privileges).

rm

Syntax: `rm file`

Example: `sudo rm backup_ls`

Purpose: Deletes files.

which

Syntax: `which command`

Example: `which ls`

Purpose: Shows location of command binary.

◆ USER MANAGEMENT

adduser

Syntax: `sudo adduser username`

Example: `sudo adduser thor`

Purpose: Interactive user creation (home + password setup).

useradd

Syntax: `sudo useradd -m username`

Example: `sudo useradd -m hulk`

Purpose: Low-level user creation without prompts.

passwd

Syntax: `sudo passwd username`

Example: `sudo passwd ironman`

Purpose: Sets or changes password.

usermod

Syntax: `sudo usermod -l new old`

Example: `sudo usermod -l tonystark ironman`

Purpose: Modifies user account.

userdel

Syntax: `sudo userdel username`

Example: `sudo userdel thor`

Purpose: Deletes user account.

◆ GROUP MANAGEMENT

groupadd

Syntax: `sudo groupadd groupname`

Example: `sudo groupadd infinity_gauntlet`

Purpose: Creates group.

groups

Syntax: `groups username`

Purpose: Shows group membership.

usermod -aG

Syntax: `sudo usermod -aG group user`

Example: `sudo usermod -aG sudo ironman`

Purpose: Adds user to group.

gpasswd

Syntax: `sudo gpasswd -d user group`

Example: `sudo gpasswd -d thanos infinity_gauntlet`

Purpose: Removes user from group.

◆ SYSTEM INFO

ps → process status
id → user info
hostname → system name
uname → system info
who → logged users
env → environment variables

◆ NETWORKING

ifconfig / ip → network config
netstat / ss → network stats
lsblk → block devices
lsusb → USB devices
lsof → open files

◆ HELP SYSTEM

man → manual pages
--help → quick help
apropos → search commands

◆ PACKAGE MANAGEMENT

dpkg -i package.deb → install deb file
apt update → refresh packages
apt install pkg → install software
apt remove pkg → uninstall
apt purge pkg → remove fully
apt upgrade → update packages
apt full-upgrade → smart upgrade

apt search → search packages
apt list → list packages
apt show → package info
apt --fix-broken install → fix dependencies
apt edit-sources → edit repo list
aptitude → interactive manager
snap install pkg → snap apps

◆ PROCESS MANAGEMENT

ps aux → all processes
pstree → process tree
pgrep → find process ID
top / htop → live monitoring
kill → terminate process
pkill → kill by name

◆ JOB CONTROL

jobs → list jobs
bg → background job
fg → foreground job
Ctrl+C → stop process
Ctrl+Z → pause process

◆ SYSTEMCTL

start / stop / restart → service control
status → check service
enable / disable → boot control
is-active → running check
is-enabled → boot check

list-units → active services
list-unit-files → all services

◆ LOGS

journalctl -xe → system logs

◆ WEB SERVERS

python3 -m http.server
php -S
npx http-server
systemctl apache2

◆ FILE MANAGEMENT

touch → create file
echo → write file
cat > file → input text
mkdir -p → nested folders
ls -l → detailed list
mv → move/rename
cp -r → copy folder
rm -rf ⚠ dangerous
rmdir → remove empty folder
tree → structure view